Asteroid Impact Damage to a Coastal City

November 8, 2021

Team 216, Problem B

Abstract

In this paper, we find the minimum mass of an asteroid required to cause substantial damage to a coastal city from an impact on an oceanic surface $1,000km$ away. With a range of asteroid masses, we investigate the final impact velocity of an asteroid when it hits the surface of the ocean with an initial starting velocity at atmospheric entry. In addition, the total energy lost due to air friction during this interval is determined to better quantify the energy efficiency of the asteroid's impact. From the impact velocity, equations describing the cavitation and resulting wave propagation in the ocean after the collision are used to find the final wave height once they reach the coastal city. When an asteroid above a certain mass impacts the ocean, the waves reaching the city amounts to the damage caused by a tsunami. Using data relating tsunami height and structural damage, the threshold for substantial damage from the asteroid impact is defined to be a wave with a height that is greater than or equal to $2m$ when reaching the city. From this threshold and the calculated wave heights (for different asteroid masses) $1,000km$ away from the impact center, we find that the minimum mass of the asteroid must be $10^9 kg$ to cause substantial damage.

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1 Problem Statement

The problem is to find the minimum mass of an asteroid required to cause substantial damage to a coastal city from an impact on the oceanic surface $1,000km$ away. From this problem, we seek to model an asteroid for a range of masses that enters Earth's atmosphere, impacts the oceanic surface, and produces oceanic waves that propagate from the impact center to the city. We find the characteristics of the waves once they reaches the coastal city, $1,000km$ from the impact center, and determine if they cause substantial damage to the city in their potential to destroy or partially destroy wooden or reinforced concrete structures.

For this problem, thermal radiation and wind currents generated from the impact are not considered for three reasons.

1. Oceanic waves are very likely to be the first to have a significant impact on the coastal city as the energy transferred to the ocean is large in comparison to that in heating or generating wind currents.

2. From the thermal conductivity of air and water, the mediums are considered to be insulating. At a distance $1,000km$ away, significant thermal changes in the city are unlikely to be made on the timescale of a day.

3. Due to the large volume of air from the center of the impact to the coastal city, wind currents generated from the impact are unlikely to have a significant effect in terms of energy due to large dissipation.

2 Assumptions

- The asteroid is perfectly spherical. This assumption is made due to the wide variability in the shapes of asteroid, leading to an over-complex model with likely insignificant changes in the final result.

- The asteroid has a uniform density distribution. This assumption is made since the density within asteroids vary but the center of mass would be approximately centered in the body.

- The asteroid impact is in the vertical direction. This assumption is made due to range of angles that the asteroid can impact the ocean surface, simplifying the computation to a vertical impact.

- The sea floor is flat from the center of the impact to the city. This assumption is made to make the model more general to different impact locations as sea floor terrains vary by location.

- The incline from the sea floor to the city is modeled vertically. This assumption is made to simplify the model without a significant decline in accuracy as the $1,000km$ distance from the impact center to the coastal city is likely to be large in comparison to sea terrain changes near the shore.

3 Asteroid Impact

3.1 Asteroid Density

We first find the density of the asteroid from previously collected data ([2]) using the line of weighted best fit average computed in MATLAB.

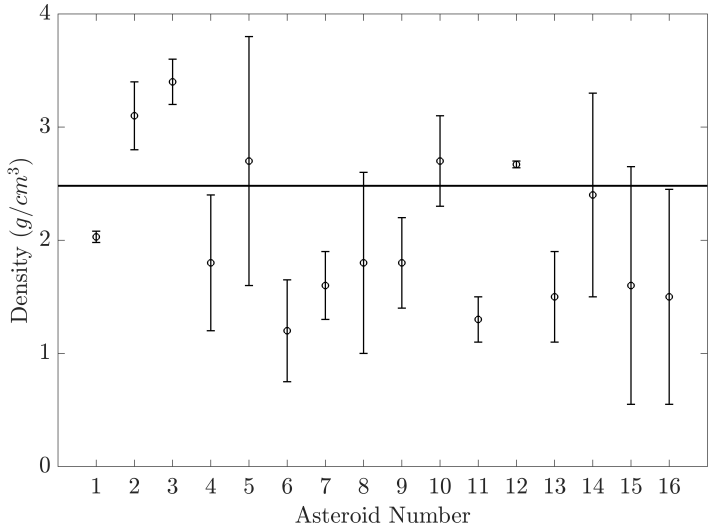


Figure 1: Asteroid density data ([2]) is collected from 16 different asteroid bodies with error bars and the weighted best fit average is found to be $2,481 kgm^{-3}$.

From the weighted best fit average on the densities of measured asteroids, we set the density of the asteroid impacting Earth as $2,481 kgm^{-3}$.

3.2 Impact Velocity

The asteroid is defined to enter Earth's atmosphere with a velocity $\mathbf{v_0}$ in the vertical $\hat{y}$ direction at point y_0. When the asteroid of a certain speed and mass enters Earth's atmosphere, the forces acting on the body are the drag force, $F_d = \frac{1}{2}\rho(y)\dot{y}^2 C_d A$, and the gravitational force, $F_g = G\frac{m_E m_A}{(y+r_E)^2}$, between the asteroid and the Earth. The variables and constants in the equation are y, the height of the asteroid from sea level; $\dot{y}$, the velocity of the asteroid; $\rho(y)$, the air density dependent on y; C_d, the drag coefficient; A, the cross-sectional area; G, the gravitational constant; m_E, the mass of the Earth; m_A, the mass of the asteroid; and r_E, the radius of the Earth.

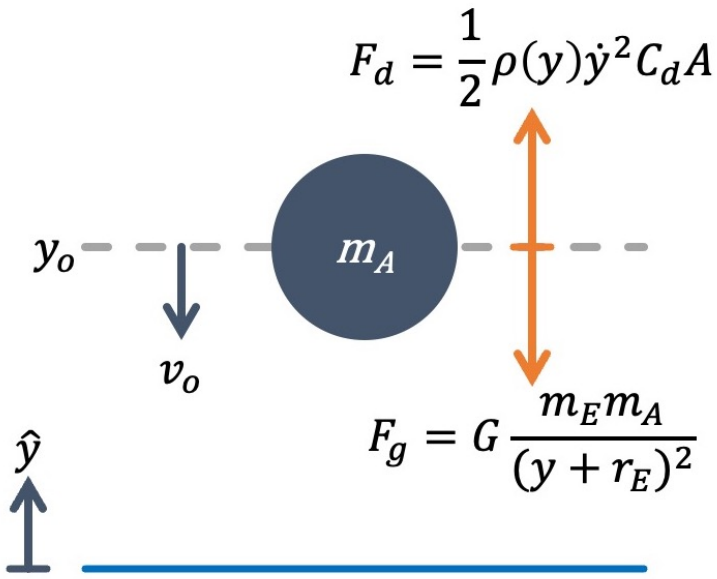


Figure 2: The forces acting on the asteroid as it enters Earth's atmosphere.

From figure 2, the governing equation of motion for the asteroid is:

$$m_A\ddot{y} = F_d - F_g \tag{1}$$

$$m_A\ddot{y} = \frac{1}{2}\rho(y)\dot{y}^2 C_d A - G\frac{m_E m_A}{(y+r_E)^2} \qquad \text{plugging in } F_d \text{ and } F_g \tag{2}$$

From the density formula with D_A as the density of the asteroid and V_A as its volume (assuming uniform density distribution on a spherical body), we solve for A in terms of m_A:

$$D_A = \frac{m_A}{V_A} \tag{3}$$

$$m_A = D_A V_A \tag{4}$$

$$= D_A\left(\frac{4}{3}\pi r_A^3\right) \qquad \text{since } V_A = \frac{4}{3}\pi r_A^3 \tag{5}$$

$$= D_A\left(\frac{4}{3}\pi\left(\frac{A}{\pi}\right)^{\frac{3}{2}}\right) \qquad \text{since } A = \pi r_A^2 \tag{6}$$

$$m_A = \frac{4}{3}\pi D_A\left(\frac{A}{\pi}\right)^{\frac{3}{2}} \qquad \text{simplifying} \tag{7}$$

$$A = \left(\frac{3m_A}{4\pi D_A}\right)^{\frac{2}{3}}\pi \qquad \text{solving for A} \tag{8}$$

Then,

$$m_A\ddot{y} = \frac{1}{2}\rho\left(y\right)\dot{y}^2 C_d \left(\frac{3m_A}{4\pi D_A}\right)^{\frac{2}{3}} \pi - G\frac{m_E m_A}{\left(y+r_E\right)^2} \qquad \text{plugging (8) into (2)} \tag{9}$$

$$\ddot{y} = \frac{\pi C_d}{2m_A}\left(\frac{3m_A}{4\pi D_A}\right)^{\frac{2}{3}} \rho\left(y\right)\dot{y}^2 - G\frac{m_E}{\left(y+r_E\right)^2} \qquad \text{rearranging} \tag{10}$$

To find the relation between the air density and height, $\rho\left(y\right)$, we use air density data from 0 to $80km$ (negligible air density past $y = 80km$ on the order of $10^{-5}kgm^{-3}$) and perform a fit function using MATLAB [3].

The result of the fit with $R^2 = 0.9975$ is:

$$\rho\left(y\right) = 1.256e^{-1.118\times10^{-4}y} \tag{11}$$

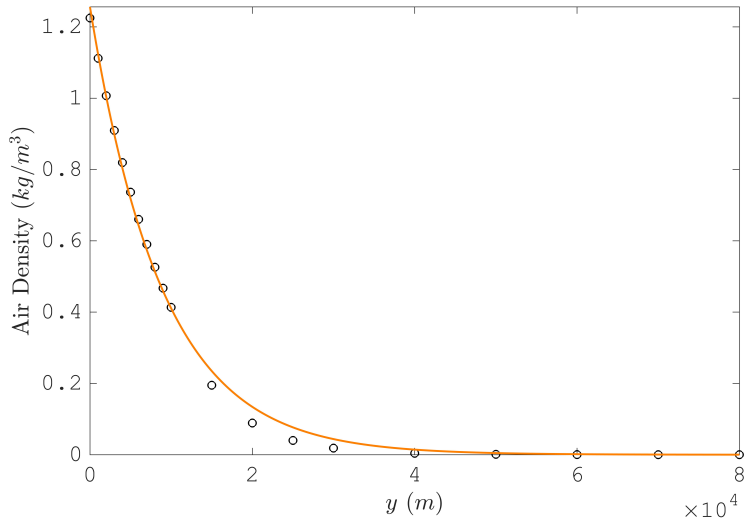


Figure 3: The air density data and curve of best fit (orange) with respect to y .

Then, plugging (11) into (10),

$$\ddot{y} = \frac{\pi C_d}{2m_A}\left(\frac{3m_A}{4\pi D_A}\right)^{\frac{2}{3}} \left(1.256e^{-1.118\times10^{-4}y}\right)\dot{y}^2 - G\frac{m_E}{\left(y+r_E\right)^2} \tag{12}$$

Plugging in the known constants $C_d = 0.877$ (for a rigid sphere in the supersonic regime [4]), $D_A = 2,481kgm^{-3}$, $G = 6.67\times10^{-11}m^3kg^{-1}s^{-2}$, $m_E = 5.97\times10^{24}kg$, and $r_E = 6.38\times10^6m$ (assuming an impact near the equator), we obtain

$$\ddot{y} = 3.6\times10^{-3}\cdot m_A^{-\frac{1}{3}}\cdot e^{-1.118\times10^{-4}y}\cdot\dot{y}^2 - \frac{3.98\times10^{14}}{\left(y+6.38\times10^6\right)^2} \tag{13}$$

The most probable asteroid encounter velocity with Earth's upper atmosphere is $v_0 = 19.3kms^{-1}$ ([5]), so with the assumption that the asteroid impacts Earth in the $\hat{y}$ direction, we have the initial condition $\mathbf{v_0} = -19.3kms^{-1}\hat{y}$. In addition, we define $y_0 = 80km\hat{y}$ as the height of atmospheric entry due to negligible air density for larger y_0 values [3].

Using these initial conditions and the governing differential equation, we numerically solve for the speed of the asteroid at impact for different m_A using the ode45 solver in MATLAB. Since the masses of asteroids are on the order of 10^9 to $10^{20}kg$ ([2]), we find the impact velocity, v_f, for masses up to this range. The results are compiled in the table below.

Mass (kg)	Impact Time (s)	Impact Velocity (ms^{-1})
10^0	220.1151	52.5338
10^2	40.0971	118.4678
10^4	5.2072	4341.1
10^6	4.3043	14005
10^8	4.1742	18043
10^{10}	4.1491	19052
10^{12}	4.1441	19278
10^{14}	4.1391	19327
10^{16}	4.1391	19337
10^{18}	4.1391	19339
10^{20}	4.1391	19340

3.3 Energy Lost on Descent

We then find the energy lost to air friction. First, we use the conservation of energy $K_i + U_i = K_f + U_f$ with the initial height $y_i = y_0$ and final height $y_f = 0$ to find the expected final impact velocity, v_e, without accounting for air resistance.

Thus,

$$K_i + U_i = K_f + U_f \tag{14}$$

$$\frac{1}{2}m_A v_0^2 + U_i = \frac{1}{2}m_A v_e^2 + U_f \tag{15}$$

We then find U_i and U_f. Since only gravitational potential energy exists, $U = U_g = G\frac{m_E m_A}{r}$, where r is the distance between the center of the asteroid and that of the Earth. To convert r to y, we note that $r = y + r_E$, so

$$U = G\frac{m_E m_A}{y + r_E} \tag{16}$$

Plugging equation (16) into (15) yields

$$\frac{1}{2}m_A v_0^2 + G\frac{m_E m_A}{y_0 + r_E} = \frac{1}{2}m_A v_e^2 + G\frac{m_E m_A}{r_E} \qquad \text{with } y_i = y_0 \text{ and } y_f = 0 \tag{17}$$

Solving for v_e,

$$v_e = \sqrt{v_0^2 - 2Gm_E\left(\frac{1}{y_0 + r_E} - \frac{1}{r_E}\right)} \tag{18}$$

$$= 19,340ms^{-1} \quad \text{plugging in the known values} \tag{19}$$

For each mass, m_A, we find the energy loss from air resistance using

$$E_{lost} = \frac{1}{2}m_A v_e^2 - \frac{1}{2}m_A v_f^2 \tag{20}$$

The results are culminated in the figure below.

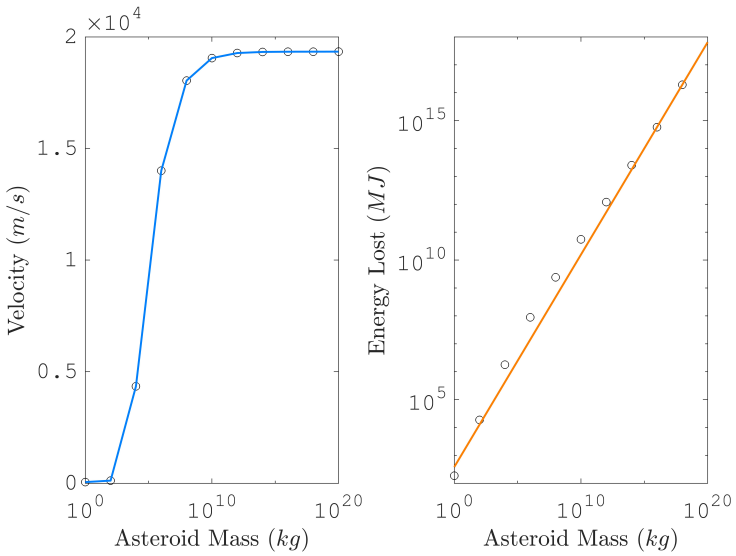


Figure 4: The impact velocity, v_f, of the asteroid (left) and the energy lost due to air friction (right). Since the speed of the largest mass in this range is equal to the speed calculated from energy conservation $19,340ms^{-1}$, a curve of best fit with $R^2 = 1$ is constructed using MATLAB for extrapolation.

From the data in figure 4, we find that the energy lost from a asteroid with a larger mass is greater than that from one with a smaller mass. However, the final impact velocity of the large asteroid is always greater. Therefore, regardless of the energy efficiency of the asteroid, a larger mass results in a larger impact velocity.

4 Oceanic Wave Propagation

Assuming that the minimum mass of an asteroid that causes substantial damage has a radius 13 times smaller than the ocean depth $6,000m$ (largest depth of the abyssopelagic zone), no shock effects arise on the sea floor and the impact is in the deep water impact (DWI) regime [1,10].

From $m_A = D_A V_A$ with $D_A = 2,481kgm^{-3}$,

$$r_A = \left(\frac{3m_A}{4\pi D_A}\right)^{\frac{1}{3}} \tag{21}$$

$$= 0.046 \cdot m_A^{\frac{1}{3}} \tag{22}$$

and the smallest asteroid mass under this radius constraint is

$$r_A \leq \frac{6000}{13} \tag{23}$$

$$0.046 \cdot m_A^{\frac{1}{3}} \leq 460 \tag{24}$$

$$m_A \leq 1.0 \times 10^{12} kg \tag{25}$$

Thus, the asteroid mass used in further calculations is $1 \times 10^{12} kg$.

4.1 Cavitation

In the deep water impact regime, the volume of the crater ([1]) formed after impact is given by

$$V = 0.75 \left(\frac{D_A}{\rho_{sea}}\right)^{\frac{1}{3}} \cdot r_A^{2.35} \cdot g^{-0.65} \cdot v_f^{1.3} \qquad \text{where } p_{sea} \text{ is the density of the ocean} \tag{26}$$

$$= 0.75 \left(\frac{D_A}{\rho_{sea}}\right)^{\frac{1}{3}} \left(\frac{3m_A}{4\pi D_A}\right)^{\frac{2.35}{3}} \cdot g^{-0.65} \cdot v_f^{1.3} \qquad \text{since } r_A = \left(\frac{3m_A}{4\pi D_A}\right)^{\frac{1}{3}} \tag{27}$$

Plugging in known constants with $p_{sea} = 1,027 kgm^{-3}$ and $g = 9.8 ms^{-2}$,

$$V = 1.6 \times 10^{-4} \cdot m_A^{0.783} \cdot v_f^{1.3} \tag{28}$$

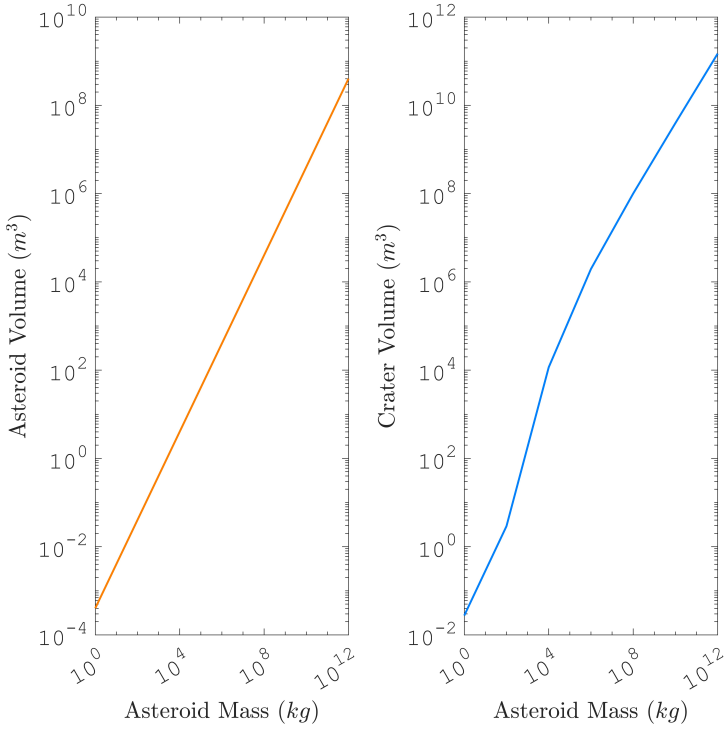


Figure 5: The asteroid volume (left) and the resulting crater volume (right) dependent on the mass of the asteroid.

The formation of this crater is accompanied by a vertical standing cone, the ejecta current, surrounding it. After collapse, the first wave, the rim wave, is formed that then propagates away from the center. In addition, gravity collapses the transient crater in the center, ultimately leading to a central peak. Its collapse results in collapse waves [1].

During this process, the collapse of the ejecta curtain paired with that of the transient crater dissipates the energy for the rim wave. Thus, the rim wave decays quickly and is negligible as the distance from the impact center increases [1].

This cycle depicted in figure 6 repeats until the sea level returns [1].

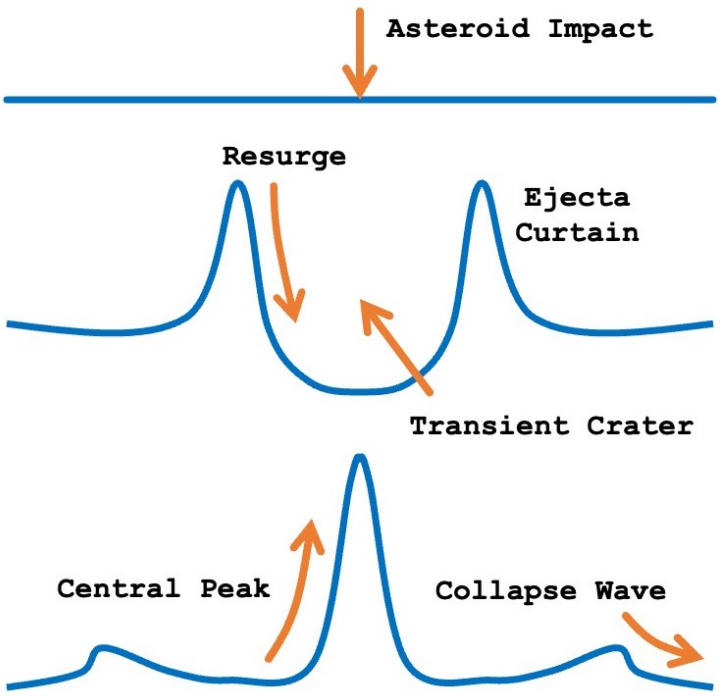


Figure 6: A depiction detailing the process of oceanic wave generation from an asteroid impact.

4.2 Wave Amplitude

Assuming that the sea floor is flat, the amplitude of these wave propagations where E is the impact energy in kilotons and r is the distance from the impact center are modeled by ([6])

$$A(r) = 9,700E^{0.5}\left(\frac{1}{r}\right) \tag{29}$$

Using the impact velocity data and $E = \frac{1}{2}m_A v_f^2$, the impact energies for the range of asteroid masses are determined and shown in the figure below.

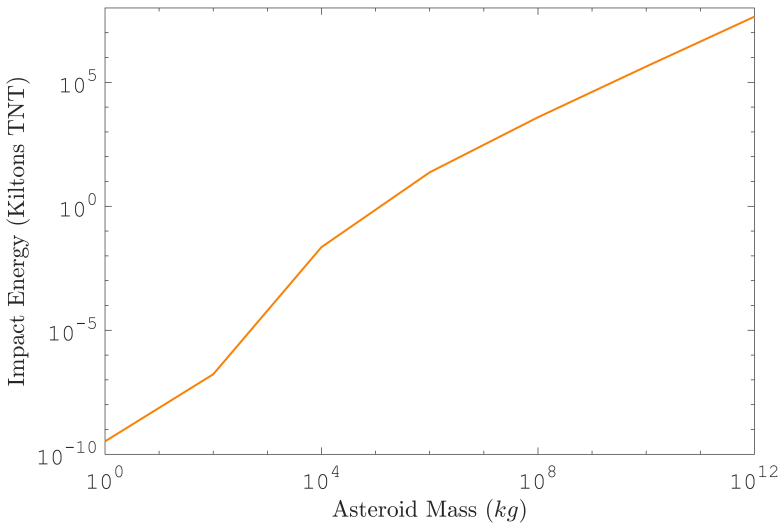


Figure 7: The impact energy in kilotons of TNT with respect to asteroid mass.

Assuming that the waves travel along a flat sea floor until they encounter a continental shelf (modeled as a vertical incline), the resulting wave heights are plotted below.

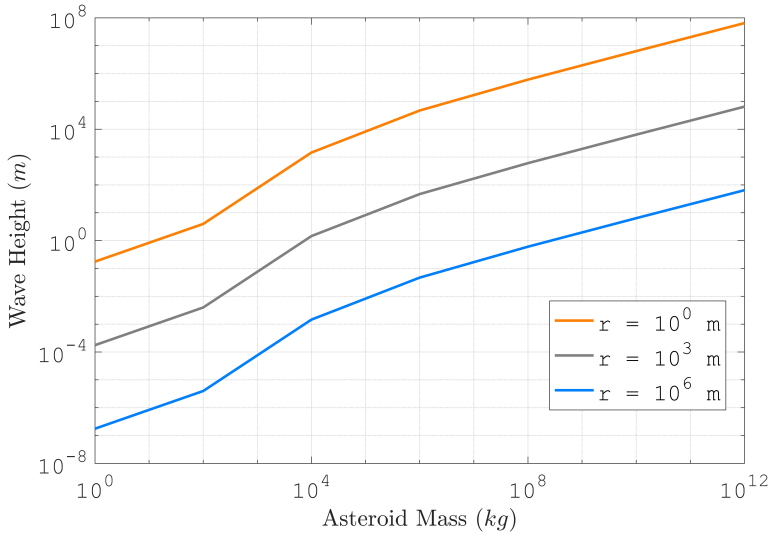


Figure 8: The wave heights are determined for an r of 10^0m, 10^3m, and 10^6m.

Due to the magnitude in the height of the propagated waves, we use tsunami data relating wave height and damage to determine the threshold for substantial damage to the coastal city. From *Tinti, 1993*, a tsunami height of $2m$ and $8m$ can cause the collapse of wooden houses and reinforced concrete structures. *Reese et al., 2007* found that a $2m$ height also causes collpase to unreinforced brick buildings. From *Ruangrassamee et al., 2006*, for a reinforced concrete building, a height of $2m$ might damage the secondary members, a $3m$ might damage the primary members, and $7m$ can completely destroy the structure.

Thus, a wave height of $2m$ is set as the cutoff for substantial damage due to its potential to destroy wooden buildings and parts of reinforced concrete structures.

From the data, setting the cutoff point at $2m$, the minimum asteroid mass to cause substantial damage to a coastal city $1,000km$ from the impact center is $10^9 kg$.

5 Discussion

The steps that were taken to reach out conclusion is split into two parts.

1. Asteroid Impact
   (a) The strengths of performing the calculations for impact velocity is that a time varying ordinary differential equation is used. It was not assumed that the velocity at the entry of the atmosphere was the the same as the velocity of the impact due to the presence of gravitational and drag forces. This led to greater accuracy in the final impact velocity determination for the range of masses used.
   (b) However, the weakness of this approach is that finding numerical solutions (MATLAB ode45 solver) to the differential equation does not give the exact solution, only an approximation.
2. Oceanic Wave Propagation
   (a) The strengths of performing the calculations for the wave amplitude is that the solutions are directly computed from the impact energy of the asteroid and no further approximations to solutions had to be made.
   (b) However, the weakness of this approach is that the wave amplitude equation is dependent on the assumption that the impact is in the deep water impact regime and the effect of the sea floor is neglected. In addition, there is a slope from the sea floor to the shore that increases the amplitude of the wave as the water depth decreases. To obtain more accurate wave heights in future work, Green's law can be used and the slope of the rise can be included in the model.

6 References

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7 Appendix

7.1 MATLAB Scripts

7.1.1 air density.m

```
% Uses air density data, uses the line of best fit (from MATLAB Curve Fitting App), and plots it

clear all
y = [1.225 1.112 1.007 0.9093 0.8194 0.7364 0.6601 0.59 0.5258 0.4671 0.4135 0.1948 0.08891 0.04008
0.01841 0.003996 0.001027 0.0003097 0.00008283 0.00001846]';
x = [0 1000 2000 3000 4000 5000 6000 7000 8000 9000 10000 15000 20000 25000 30000 40000 50000 60000
70000 80000]';
x_values = linspace(min(x),max(x),1000);
fit_result = @(x) 1.256*exp(-0.0001118*x);
plot(x,y,'ko');
hold on
plot(x_values,feval(fit_result,x_values),'Color',[1,0.5,0],'LineWidth',1.5)
hold off
ylim([0 inf])
ylabel('Air Density ($kg/m^3$)','Interpreter','latex','FontSize',12)
xlabel('$y$ ($m$)','Interpreter','latex','FontSize',12)
```

7.1.2 asteroid density.m

```
% Uses asteroid density data, creates the line of best fit average, and plots it

clear all
x = [1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16]';
y = [2.03 3.1 3.4 1.8 2.7 1.2 1.6 1.8 1.8 2.7 1.3 2.67 1.5 2.4 1.6 1.5]';
error = [0.05 0.3 0.2 0.6 1.1 0.45 0.3 0.8 0.4 0.4 0.2 0.03 0.4 0.9 1.05 0.95]';
weights = error.^(-2);

x_values = [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17]';

figure
errorbar(x,y,error,'ko','MarkerSize',5,'LineWidth',1);
hold on

fit_function = @(b,x) b+x.*0;

[fit_horizontal,goodness] = fit(x,y,fit_function,'Weights',weights);

disp(fit_horizontal)

plot(x_values,feval(fit_horizontal,x_values),'Color','k','LineWidth',1.5)

xlim([0 17])
xticks([1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16])
ylim([0 4])
yticks([0 1 2 3 4])
xlabel('Asteroid Number','Interpreter','latex','FontSize',12)
ylabel('Density ($g/cm^3$)','Interpreter','latex','FontSize',12)
```

7.1.3 crater volume.m

```
% Uses mass and impact velocity data to create plots of asteroid volume and
% crater volume vs. asteroid mass

clear all
x = [1E+00 1E+02 1E+04 1E+06 1E+08 1E+10 1E+12]';
y = [52.5338 118.4678 4341.1 14005 18043 19052 19278]';
volume = x/2481;
V = (1.6*10^(-4)).*(x.^0.783).*y.^1.3;

figure
subplot(1,2,1)
loglog(x,volume,'Color',[1,0.5,0],'LineWidth',1.5);
ylabel('Asteroid Volume ($m^3$)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)
xlim([1,1E+12]);
xticks([1E+00 1E+04 1E+08 1E+12]);

subplot(1,2,2)
loglog(x,V,'Color',[0,0.5,1],'LineWidth',1.5);
ylabel('Crater Volume ($m^3$)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)
xlim([1,1E+12]);
xticks([1E+00 1E+04 1E+08 1E+12]);
```

7.1.4 cumulative impact data.m

```
% Uses mass and impact velocity data to find energy loss and create plots
% of impact velocity and energy lost vs. asteroid mass

clear all
x = [1E+00 1E+02 1E+04 1E+06 1E+08 1E+10 1E+12 1E+14 1E+16 1E+18 1E+20]';
y = [52.5338 118.4678 4341.1 14005 18043 19052 19278 19327 19337 19339 19340]';
energy_difference = 0.5.*x*19340^2 - 0.5.*x.*y.^2;
x_energy = [1E+00 1E+02 1E+04 1E+06 1E+08 1E+10 1E+12 1E+14 1E+16 1E+18]';
energy_difference_values = energy_difference(1:end-1);

figure
subplot(1,2,1)
semilogx(x,y,'ko');
hold on
semilogx(x,y,'Color',[0,0.5,1],'LineWidth',1.5);
hold off
ylabel('Velocity ($m/s$)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)

subplot(1,2,2)
loglog(x,energy_difference./1000000,'ko');
hold on
loglog(x,((3.832*10^8).*x.^0.7613)/1000000,'Color',[1,0.5,0],'LineWidth',1.5)
hold off
ylabel('Energy Lost ($MJ$)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)
```

7.1.5 impact energy.m

```
% Calculates impact energy from mass and impact velocity and plots it

clear all
x = [1E+00 1E+02 1E+04 1E+06 1E+08 1E+10 1E+12]';
y = [52.5338 118.4678 4341.1 14005 18043 19052 19278]';
energy = (0.5.*x.*y.^2)*2.39006E-13;

figure
loglog(x,energy,'Color',[1,0.5,0],'LineWidth',1.5);
ylabel('Impact Energy (Kiltons TNT)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)
xlim([1,1E+12]);
xticks([1E+00 1E+04 1E+08 1E+12]);
```

7.1.6 impact velocity.m

```
% Uses the ode45 solver to solve the differential equation to find position
% and velocity vs. time

clear all
m = 10^2;
t_max = 41;
t_interval = linspace(0,t_max,1000);
syms y(t)
[V] = odeToVectorField(diff(y,2) == (3.6*10^(-3))*(m^(-1/3))*exp(-1.118*10^(-4)*y)*diff(y)^2-
(3.98*10^14)/(y+6.38*10^6)^2)

M = matlabFunction(V,'vars', {'t','Y'})
options=odeset('RelTol',1e-8,'AbsTol',1e-8);
sol = ode45(M,[0,max(t_interval)],[80000 -19300],options);

figure
subplot(1,2,1)
position = deval(sol,t_interval,1);
plot(t_interval,position,'Color',[0,0.5,1],'LineWidth',2)
xlabel('t (s)','Interpreter','latex','FontSize',12)
ylabel('Position (m)','Interpreter','latex','FontSize',12)
ylim([0,inf])

subplot(1,2,2)
velocity = deval(sol,t_interval,2);
plot(t_interval,abs(velocity),'Color',[1,0.5,0],'LineWidth',2)
xlabel('t (s)','Interpreter','latex','FontSize',12)
ylabel('Speed ($m/s$)','Interpreter','latex','FontSize',12)

[~,time_index] = min(abs(position));
time = t_interval(time_index)
speed = abs(velocity(time_index))
```

7.1.7 wave amplitude.m

```
% Calculates wave amplitude from given data and creates plots of wave
% amplitude vs. asteroid mass

clear all
r = [10^0,10^3,10^6];
x = [1E+00 1E+02 1E+04 1E+06 1E+08 1E+10 1E+12]';
y = [52.5338 118.4678 4341.1 14005 18043 19052 19278]';
energy = (0.5.*x.*y.^2)*2.39006E-13;
amplitudes = [];
for i = 1:3
    amplitude_new = 9700.*(energy.^0.5)*1/r(i);
    amplitudes = [amplitudes,amplitude_new];
end
amplitudes = amplitudes';

figure
loglog(x,amplitudes(1,:),'Color',[1,0.5,0],'LineWidth',2);
hold on
loglog(x,amplitudes(2,:),'Color',[0.5,0.5,0.5],'LineWidth',2);
loglog(x,amplitudes(3,:),'Color',[0,0.5,1],'LineWidth',2);
hold off
ylabel('Wave Height ($m$)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)
legend('r = 10^0 m','r = 10^3 m','r = 10^6 m')
xlim([1,1E+12]);
xticks([1E+00 1E+04 1E+08 1E+12]);
yticks([1E-08 1E-04 1E+00 1E+04 1E+08]);
grid on

figure
loglog(x,amplitudes(3,:),'Color',[0,0.5,1],'LineWidth',2);
hold on
loglog(x,zeros(1,length(x))+2,'Color','red','LineWidth',2);
hold off
ylabel('Wave Height ($m$)','Interpreter','latex','FontSize',12)
xlabel('Asteroid Mass ($kg$)','Interpreter','latex','FontSize',12)
xlim([1,1E+12]);
xticks([1E+00 1E+04 1E+08 1E+12]);
yticks([1E-08 1E-04 1E+00 1E+04]);
grid on
```